Where Do Text Embeddings Keep Geography? Layer-wise Probes and Map Views of Base vs. Retrieval-Tuned Models

Team N: Artem Frenk (agf6353), Nathan Chen (nsc9305), Tian-hao Zhang (tz3147), Kenta Ploch (kp4197)
github.com/afrenkai/ComeALittleCloser

Progress since the proposal

- **Probing pipeline (complete).** Ridge probes on every layer's pooled hidden state for 34,196 GeoNames cities and 11,833 radio stations; 10° spatial-block and random splits, 3 seeds, block-bootstrap CIs, a training-only country-centroid control, and a held-out test split that has not been touched. Pinned revisions, data checksums and every cluster job are logged in the repository.
- **Models.** 18 checkpoints: four decoder *base→retrieval* pairs whose model cards name the base (Qwen3 0.6B/4B/8B; Mistral-7B→e5-mistral and Gemma-2-9B→bge-multilingual-gemma2, running now), XLM-R base/large with E5 and the BGE-M3 training stages, and two random-initialization controls.
- **Validation results (14 models; test not yet run).** A *country-residual* probe (offset from the country centroid) shows within-country geography that grows with scale: 12 / 24, 120 / 152, and 175 / 201 km gain over the country centroid for Qwen3 0.6B, 4B, 8B (base / retrieval). Encoders stay near 0. Retrieval tuning does not remove this information, and the 8B retrieval vector itself beats the country centroid by 311 km [89, 565].
- **Visualization.** The globe explorer (origin, genre filter, true-nearest vs. model rankings, 3-D embedding view, audio) and static depth-curve and distortion-arrow figures exist.

Response to feedback

1. **Visualization contribution.** Three linked views. *(V1) Globe ranking:* model top-k vs. true k nearest drawn as lines from the origin; shows picks that stop at borders or jump to same-language regions, which mean regret hides. *(V2) Layer-by-region map* (new): 10° blocks colored by probe gain over the country centroid, with a layer slider linked to the depth curve and a base−retrieval difference mode; shows *where* and *at which layer* geography appears or is lost, which the global average hides. *(V3) Distortion map:* true→predicted arrows per city, exposing pulls toward capitals or continent centers. Evaluation (no IRB): pre-stated case studies, each confirmed quantitatively (e.g. per-region gain spread, border-crossing rate), plus a think-aloud expert walkthrough with 2–3 NLP/GIS researchers.
2. **Critique of key papers.** *Gurnee & Tegmark* probe base LMs with R^2 on raw latitude/longitude. R^2 rewards knowing the country: on our data, the country centroid alone reaches $R^2 = 0.95$ (latitude) and a *randomly initialized* network 0.78 when the text names the country. Their R^2 does not separate country identity from within-country position, and they probe base LMs only, not retrieval embeddings or what retrieval training keeps. *Decoupes et al.* compare cosine and geographic distance on final embeddings only, so absent geography cannot be told apart from geography present in earlier layers but unused; their vocabulary analysis is an indirect proxy. *Gap:* geography beyond the country, by layer and region, before vs. after retrieval training.
3. **Scope.** Core: (i) layer probes, base vs. retrieval; (ii) retrieval proximity metrics; (iii) V1–V3 with case studies. Stretch: audio pilot and SoundingEarth with LLM re-captioning, learned adapter. Dropped: H3 adjacency (10° blocks suffice).
4. **Subspace amplification** is dropped. Coordinate perturbation stays as a secondary test: it shows whether rankings respond to coordinate *text*, but cannot say *why* explicit coordinates fail (tokenization, no numeric geometry, or genre dominance).
5. **Metrics and framing.** Per (origin, genre) query over the genre-eligible pool: Spearman ρ between cosine and Haversine ranks, overlap@10 with the 10 nearest, and excess mean distance (km) of the top 10, against a random ordering of the same pool (1,000 permutations) and the Haversine sort. We adopt the suggested framing, *which notion of proximity embeddings encode*, and add same-country and same-language proximity as alternative references.
6. **Small fixes.** New title; models counted as above (Jina dropped; EmbeddingGemma stays in retrieval only, as it has no released base); typo fixed; SoundingEarth ground truth is each clip's recorded coordinates (stretch only).

Plan and division of labor

- **Owners.** Probing: Tian-hao Zhang. Interface V1–V3: Artem Frenk. Retrieval metrics and baselines: Nathan Chen. Case studies, expert walkthrough, audio stretch: Kenta Ploch.
- **Nov 4–10:** freeze test cells; run the held-out test once; export per-region scores. **Nov 11–24:** retrieval metrics with random baselines; build V2 and link it to V1. **Nov 25–Dec 6:** case studies and expert walkthrough. **Dec 7–20:** paper; stretch goals only if the core is done.